

AEM-X100 — Error Code Reference

Equipment: AEM-X100 Wafer Handling System

Document ID: SYN-X100-ERROR_CODE_REFERENCE **Revision:** A **Status:** Synthetic Training Data

IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Provide a synthetic alarm catalogue for retrieval, classification, and diagnostic evaluation.

1. Alarm Catalogue

Code	Description	Typical area	Initial documented action
E101	Wafer presence not detected	WPS / loading	Check WPS status and loading conditions
E102	Wafer presence signal unstable	WPS	Check sensor condition and repeat only the approved check
E201	Transfer position not confirmed	WTM	Check position-sensor evidence; escalate if persistent
E202	Transfer motion timeout	WTM	Stop repeated commands and escalate
E301	Safety interlock active	SIM	Stop operation and follow Safety Procedure
E302	Emergency stop active	SIM	Do not continue operation; follow site safety/escalation process
E401	Vacuum handling condition not achieved	VHM	Follow documented VHM check; escalate if unresolved
E501	SCU communication fault	SCU	Record alarm and follow escalation process

2. Alarm Handling Rules

- Always capture the exact alarm code before troubleshooting.
- Do not infer an alarm code from a symptom when the SCU code is available.
- If multiple alarms are present, prioritize safety-interlock and emergency-stop conditions.
- If an alarm does not appear in this reference, the assistant should state that no matching evidence was found.

3. Diagnostic Examples

- E101 should normally route to Sensor Troubleshooting and may also require review of Wafer Loading Procedure.
- E301 should route to Safety Procedure before any diagnostic action.
- E202 should route to Diagnostic reasoning and then Escalation Procedure if the documented checks do not resolve the condition.

End of synthetic document.